Course Plan and Policies

EM-614: Advanced Probability and Statistics
Module-I, Jun-Sep 2024
Indian Institute of Technology Gandhinagar

Instructors:	**Teaching Assistant**
Raj Srinivasan Office: University of Saskatchewan, Canada Email: raj@math.usask.ca raj.srinivasan@iitgn.ac.in	Mohit Gupta Email: mohit.g@iitgn.ac.in

Class Location and Meeting Times

Sessions:
Friday, Timings 8:30 PM to 10:00 PM (IST)
Saturday: Timings 09:00AM to 10:30AM (IST)
Mode of Engagement:
Online on Zoom. Sign-in using your IITGN email ID to connect [link]

Reference Texts

- Schervish, Mark J., and Morris H. DeGroot. *Probability and Statistics*. Vol. 563. London, UK: Pearson Education, 2014.
- Casella, George, and Roger L. Berger. *Statistical inference*. Cengage Learning, 2021

Course Website

URL: https://sites.google.com/iitgn.ac.in/dsdm-em-614
The course will be fully managed though the website and all the major announcements, lecture notes, scheduled activities will be uploaded to the website only. Students are advised to check the website regularly for any updates regarding the course.

Course Objectives

The aim of the course is to enable students to develop an understanding of probability theory, hypothesis testing, linear models, stochastic processes, joint distributions, and copula, enabling them to apply these tools effectively across various domains.

Grading (To be revisited upon starting of the class)

The final grade shall be awarded based on the total score of the student that will be calculated using the following weights.

- Mid terms : 30%
- Assignments: 30%
- Examination-I: 40%

It should be noted that the above distribution is tentative and can be modified during the module by the instructor based on student feedback. Changes, if any, shall be adequately discussed in the class and a consensus between the instructor and the students will be obtained, before implementation.

Mid Terms

There will be two midterm tests; the dates will be announced later.

Assignments

There will be several assignments in the course. The questions in the assignment will be based on the concepts covered in the course. Students are encouraged to discuss the problems with peers, TAs and instructors before submission. The following points must be noted:

- All the assignments will have equal weights, irrespective of its maximum marks.
- Some of the questions in the assignments might carry extra credits. These marks will be scaled and added directly to the total score that the student receives in the course.

Submission of Assignments

- All the assignments must be submitted using the Online Forms link given in the assignment.
- Assignments submitted via any other means will not be accepted.

Examinations

The assessments for Examinations shall be done through an online exam. The students shall be given a question paper and they will be required to upload their answers during a specified time.. The grades shall be awarded to each student, which shall be converted into marks using the adjoining table:

Letter Grade	A+	A	A-	B	B-	C	C-	D	F
Grade Point	11	10	9	8	7	6	5	4	0

EM-614: Advanced Probability and Statistics
Instructor: Raj Srinivasan
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Week No.	Start Date	End Date	L/T No.	L/T Date	Day	Contents
W1	17-Jun-24	23-Jun-24	L-01	21-Jun-24	Friday	Course Overview and Policies **Probability Theory** Foundations of Probability: Axioms
			L-02	22-Jun-24	Saturday	Conditional probability, independence, Bayes' theorem.
W2	24-Jun-24	30-Jun-24	L-03	28-Jun-24	Friday	Random Variables and Distributions: Discrete probability distributions.
			L-04	29-Jun-24	Saturday	Continuous random variables, probability density functions.
W3	01-Jul-24	07-Jul-24	L-05	05-Jul-24	Friday	Moments and Characteristic Functions: Expectation, variance, skewness, kurtosis, and moment-generating functions.
			L-06	06-Jul-24	Saturday	Limit Theorems: Law of large numbers, central limit theorem, convergence concepts
W4	08-Jul-24	14-Jul-24	L-07	12-Jul-24	Friday	**Estimation Theory** Point Estimation: Maximum likelihood estimation, method of moments, Bayesian estimation. Interval Estimation: Confidence intervals, prediction intervals, and tolerance intervals.
			L-08	13-Jul-24	Saturday	Properties of Estimators: Unbiasedness, efficiency, consistency, and relative efficiency. Asymptotic Analysis: Large sample theory, asymptotic normality, delta method.
W5	15-Jul-24	21-Jul-24	L-09	19-Jul-24	Friday	**Hypothesis Testing** Formulating Hypotheses: Null and alternative hypotheses, types of errors, significance levels.
			L-10	20-Jul-24	Saturday	Test Statistics and p-values: t-tests, chi-square tests, F-tests, likelihood ratio tests. **Linear Models and Regression Analysis** Simple and Multiple Linear Regression: Assumptions, estimation, inference, diagnostics.
W6	22-Jul-24	28-Jul-24	L-11	26-Jul-24	Friday	**Hypothesis Testing:** Multiple Comparisons: Bonferroni, Tukey's HSD, Holm's method.
			L-12	27-Jul-24	Saturday	Nonparametric Testing: Rank-based tests, bootstrapping, permutation tests.
W7	29-Jul-24	04-Aug-24	L-13	02-Aug-24	Friday	**Linear Models and Regression Analysis** Model Selection: AIC, BIC, cross-validation, regularization (Lasso, Ridge). Simple and Multiple Linear Regression: Assumptions, estimation, inference, diagnostics Generalized Linear Models (GLM): Logistic regression, Poisson regression, link functions. Multivariate Analysis: Principal component analysis, factor analysis, canonical correlation.
			L-14	03-Aug-24	Saturday	**Introduction to Stochastic Processes** Classification of Stochastic Processes: Markov chains, Poisson processes, Brownian motion.
W8	05-Aug-24	11-Aug-24	L-15	09-Aug-24	Friday	Stationary Processes: Weak and strong stationarity, autocovariance functions, spectral analysis.
			L-16	10-Aug-24	Saturday	Applications in Finance, Engineering, Environmental Science: Modelling real-world phenomena.

Week No.	Start Date	End Date	L/T No.	L/T Date	Day	Contents
W9	12-Aug-24	18-Aug-24	L-17	16-Aug-24	Friday	**Joint Distributions and Copula** Bivariate and Multivariate Distributions: Joint, marginal, and conditional distributions.
			L-18	17-Aug-24	Saturday	Copulas: Definitions, properties, Sklar's theorem, common copula families (Gaussian, Clayton, Gumbel).
W10	19-Aug-24	25-Aug-24	L-19	23-Aug-24	Friday	Dependence Measures: Correlation, rank correlation, tail dependence.
			L-20	24-Aug-24	Saturday	Applications in Risk Management, Finance, Climate Studies: Modelling complex dependencies.
W11	26-Aug-24	01-Sep-24	**ONLY Evaluations: No Lectures and/or Tutorials**			